



Chapter 8

Depreciation and Taxes

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Depreciation and Taxes

(5 hours)

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Definition

Depreciation is the systematic allocation of a tangible asset's cost over its estimated useful life. It is the gradual decrease in the value of an asset over time due to usage, age, wear and tear, or obsolescence

Causes of Depreciation

The causes are generally categorized into physical and functional factors:

1. *Physical Depreciation*: The actual wearing out or deterioration of the asset's physical components.
 - Wear and Tear: Deterioration resulting from regular use in business operations (e.g., friction in moving parts, abrasion, or fatigue).
 - Deterioration: Gradual decay, corrosion (rusting), or erosion due to environmental exposure (e.g., rotting wood or cracking plastic).

Definition

- Accidental Damage: Sudden and unexpected physical loss caused by events like fire, floods, or earthquakes.
- 2. *Functional Depreciation*: The loss of value due to a decrease in demand for the asset's specific function.
- Obsolescence: Occurs when an asset becomes "out-of-date" because newer, more efficient, or more economical technology becomes available.
- Inadequacy: When an asset can no longer meet the current or anticipated demand of the business, even if it is still in good working condition.
- Changes in Style/Design: A shift in consumer taste or market standards that makes the goods produced by the asset less desirable.

Definition

Other Causes:

- **Effluxion of Time:** The simple passage of time, especially for assets with limited legal lives like leasehold properties or patents.
- **Depletion:** The exhaustion of natural resources like minerals, oil, or timber through extraction.
- **Economic Factors:** Changes in the market such as reduced demand for the product, general business recessions, or zoning changes.

Definition

Depreciation is the systematic recognition of the economic value of an asset that has been consumed in generating revenue over time.

Depreciation signifies:

- Allocation of cost
- Consumption of asset utility
- Preservation of capital
- Realistic financial reporting

It is a bridge between : Engineering usage and Financial accountability.

Definition

Depreciation is more than just a bookkeeping entry; it is a critical tool for financial analysis and strategic planning. Its importance stems from its impact on a firm's cash flow, **tax obligations**, and long-term decision-making.

Core Importance of Depreciation

- 1) **Tax Shield and Savings:** Depreciation is a non-cash expense that is subtracted from revenue before calculating taxes. This reduces taxable income, effectively lowering the amount of tax a company owes and increasing available after-tax cash flow.
- 2) **Cost Recovery:** It allows a business to systematically recover the initial investment of a large capital asset over its useful life, rather than taking a massive financial hit in the year of purchase.
- 3) **Accurate Financial Reporting:** By applying the matching principle, depreciation ensures that the cost of an asset is recognized in the same periods as the revenue it helps generate. This provides a more realistic picture of annual profitability and the current book value of assets.

Definition

- 4) **Replacement Planning:** Tracking an asset's declining value helps engineers and managers anticipate when equipment will become obsolete or too costly to maintain, facilitating timely budgeting for replacements.
- 5) **Project Feasibility (NPV & IRR):** In engineering economic analysis, depreciation is a key variable in calculating Net Present Value (NPV) and Internal Rate of Return (IRR), as it significantly alters the timing and magnitude of after-tax cash flows.
- 6) **Asset Management and Valuation:** Accurate records of accumulated depreciation help a company monitor asset performance and provide necessary data for insurance, securing loans, or selling business assets.

Definition

If Depreciation Is Not Shown Properly:

- *Profit appears higher than actual*

Without depreciation, expenses look smaller and profit is overstated.

- Company may pay higher corporate tax

Higher reported profit leads to higher tax payment than necessary.

- *Asset value becomes unrealistic*

Equipment /Asset is shown at higher value even after years of use.

- *Financial statements become misleading*

Investors, banks, and auditors get incorrect financial information.

- *Audit problems may arise*

Improper depreciation can lead to questioning of financial reports.

- *Wrong management decisions may be taken*

Owners may overestimate profit and delay replacement planning.

Depreciation calculation

Following are the primary methods used to calculate depreciation:

- i. Straight-Line Method (SLM)
- ii. Declining Balance Method (DBM) (also known as the Reducing Balance or Written Down Value method)
- iii. Double Declining Balance Method (DDB)
- iv. Sum-of-the-Years'-Digits Method (SYD)
- v. Units of Production Method (also known as the Service Output Method)
- vi. Sinking Fund Method
- vii. Annuity Method
- viii. Modified Accelerated Cost Recovery System (MACRS) (primarily used for tax purposes in the US)

Straight Line Method

The Straight Line Method is a depreciation method in which the asset loses equal value every year throughout its useful life.

To calculate the annual depreciation, subtract the salvage value (the estimated value at the end of its life) from the initial cost and divide it by the number of years in the useful life:

$$D = \frac{P - S}{n}$$

Where:

P = First cost

S = Salvage value

n = Useful life (years)

D = Annual depreciation

Straight Line Method

Key Characteristics

- a) **Constant Rate:** The depreciation expense is the same for every year of the asset's life.
- b) **Linear Decrease:** If you were to graph the Book Value (the remaining value of the asset) over time, it would appear as a straight line sloping downward.
- c) **Simplicity:** It is very easy to calculate and understand, making it the standard for general financial reporting.

Book Value (BV):

It refers to the net value of an asset or an entire company as recorded on financial statements. It represents the historical cost of an investment adjusted for any cumulative changes, such as depreciation

Straight Line Method

Formula:

$$\text{Book Value}_k = \text{Initial Cost} - \text{Accumulated Depreciation up to year } k$$

Purpose: It tracks the remaining unrecovered investment in an asset. In engineering projects, this value is used to determine taxable gains or losses if the asset is sold before the end of its useful life.

Final Value: At the end of an asset's useful life, the book value typically equals its estimated salvage value.

Straight Line Method

If Rate is Given

Annual Depreciation

$$= \text{Total depreciable value} \times \frac{\text{Rate (\%)}}{100}$$

Rate of Depreciation

$$= \frac{\text{Amount of annual depreciation}}{\text{Total depreciable value}} \times 100$$

Straight Line Method

Alternatively,

$$\text{Rate of depreciation} = \frac{1}{N} \times 100$$

(where N = life of asset)

Note:

Total depreciable value = Initial cost + All other expenses – Salvage value

Declining balance Method

The **Declining Balance Method** (also known as the Reducing Balance or Diminishing Balance method) is an accelerated depreciation technique. It applies a constant percentage to the asset's remaining book value each year, resulting in much higher depreciation charges in the early years of an asset's life.

Unlike the straight-line method, which uses the initial cost, this method calculates depreciation based on the Book Value (Initial Cost - Accumulated Depreciation) at the start of each period. As the book value "declines" every year, the actual dollar amount of depreciation also decreases

The **Double Declining Balance (DDB)** method is an accelerated depreciation method. It reduces the asset's value very fast in the early years, then depreciation gradually becomes smaller

Declining balance Method

When salvage value is given:

$$d = 1 - \left(\frac{S}{P} \right)^{1/n}$$

Where:

- S = Salvage value
- P = Initial cost
- n = Useful life

Double Declining Balance (DDB)

$$d = \frac{2}{n}$$

$$D_t = \frac{2}{n} \times BV_{t-1}$$

Declining balance Method

Annual Depreciation Formula

$$D_t = d \times BV_{t-1}$$

Where:

- D_t = Depreciation in year t
- d = Depreciation rate (decimal form)
- BV_{t-1} = Book value at beginning of year t

Double Declining Balance (DDB)

$$D_t = \frac{2}{n} \times BV_{t-1}$$

Here, rate = $2 \times$ Straight Line rate.

Declining balance Method

➤ Consider the following accounting information for a computer system:

Initial cost of asset = Rs. 10,000

Useful life = 5 years

Compute the annual depreciation amount and resulting book value using the Double Rate Declining Balance Method.

Given:

Original cost (B) = Rs. 10,000

Useful life (N) = 5 years

Calculate Declining Balance Rate

$$R = \frac{1}{N} \times 100 \times 2$$

$$R = \frac{1}{5} \times 100 \times 2$$

$$R = 40\%$$

Declining balance Method

Annual Depreciation and Book Value Calculation

Year (K)	Beginning Book Value (Rs.)	Annual Depreciation @ 40% (Rs.)	Ending Book Value (Rs.)
1	10,000	4,000	6,000
2	6,000	2,400	3,600
3	3,600	1,440	2,160
4	2,160	864	1,296
5	1,296	518.4	777.6

Depreciation decreases each year because 40% is applied to the remaining book value, not the original cost.

This is why the method is called **Double Declining Balance**

Declining balance Method

- Compute the annual depreciation allowances and the resulting book value using the double declining balance method with switching over to straight line method.

Cost of asset = Rs. 1,00,000

Useful life = 5 years

Salvage value = Rs. 20,000

Solution

$$I = 1,00,000$$

$$N = 5 \text{ years}$$

$$S = 20,000$$

Depreciation by Straight Line Method

$$\begin{aligned} D_{SL} &= \frac{I - S}{N} \\ &= \frac{1,00,000 - 20,000}{5} = \frac{80,000}{5} = 16,000 \end{aligned}$$

Declining balance Method

Declining Balance Rate

$$\begin{aligned}d &= 1 - \sqrt[N]{\frac{S}{I}} \\&= 1 - \sqrt[5]{\frac{20,000}{1,00,000}} \\&= 1 - \sqrt[5]{0.2} = 0.27522 = 27.522\%\end{aligned}$$

Declining balance Method

Depreciation Calculation

(Switchover from 27.522% declining balance to straight line method)

Year	Beginning Book Value	Declining Balance @ 27.522%	Compare	Straight Line Method	Depreciation Selected
1	1,00,000	27,522	>	16,000	27,522
2	72,478	19,947	>	13,120	19,947
3	52,531	14,458	>	10,843	14,458
4	38,073	10,478	>	9,036	10,478
5	27,595	7,595	=	7,595	7,595 (Switch)

Declining balance Method

Beginning book value = 72,478

Remaining life = 4 years

$$SL_2 = \frac{72,478 - 20,000}{4} = \frac{52,478}{4} = 13,120$$

Year 3

Beginning book value = 52,531

Remaining life = 3 years

$$SL_3 = \frac{52,531 - 20,000}{3} = \frac{32,531}{3} = 10,843$$

Year 4

Beginning book value = 38,073

Remaining life = 2 years

$$SL_4 = \frac{38,073 - 20,000}{2} = \frac{18,073}{2} = 9,036$$

Declining balance Method

Total Depreciation

$$= 27,522 + 19,947 + 14,458 + 10,478 + 7,595$$

$$= 80,000$$

$$\text{Final Book Value} = 1,00,000 - 80,000 = 20,000 \text{ (Salvage Value)}$$

Sum of the year digit method

The SOYD method is an accelerated depreciation method in which the annual depreciation is proportional to the remaining life of the asset.

It allocates higher depreciation in the early years and decreases uniformly each year.

The Sum-of-Years'-Digits method is an accelerated depreciation method in which annual depreciation is determined by multiplying the depreciable amount by a fraction whose numerator is the remaining life and denominator is the sum of the years' digits.

Sum of Years' Digits (SYD)

$$SYD = \frac{n(n+1)}{2}$$

Sum of the year digit method

- A machine has a cost of Rs. 36,000 and an estimated salvage value at the end of the 5th year of Rs. 6,000. Calculate the annual depreciation charges under the Sum of Years' Digits (SOYD) Method.

Given:

- Initial cost, = Rs. 36,000
- Salvage value, $S = \text{Rs. } 6,000$
- Useful life, $N = 5$ years

Depreciation amount each year

$$36,000 - 6,000 = 30,000$$

Calculate Sum of Years' Digits (SYD)

$$\text{SYD} = 5 + 4 + 3 + 2 + 1 = 15$$

$$\begin{aligned}\text{SYD} &= \frac{N(N+1)}{2} \\ &= \frac{5(5+1)}{2} = \frac{5 \times 6}{2} = 15\end{aligned}$$

Sum of the year digit method

Year	SOYD Depreciation
1	$\frac{5}{15} \times 30,000 = 10,000$
2	$\frac{4}{15} \times 30,000 = 8,000$
3	$\frac{3}{15} \times 30,000 = 6,000$
4	$\frac{2}{15} \times 30,000 = 4,000$
5	$\frac{1}{15} \times 30,000 = 2,000$

Sinking Fund Method

A **sinking fund** is a dedicated, interest-bearing account created by setting aside periodic, systematic payments to accumulate a specific sum for future capital expenditures, such as replacing machinery, equipment, or major infrastructure at the end of its useful life. It serves as a financial strategy to manage large, known, future costs, often used for debt retirement.

So depreciation is based on:

- Annual deposit, and
- Interest earned on deposits

Standard Formula (Annual Depreciation)

$$D = (P - S) (A/F, i, n)$$

Where:

$$(A/F, i, n) = \frac{i}{(1 + i)^n - 1}$$

So,

$$D = (P - S) \left[\frac{i}{(1 + i)^n - 1} \right]$$

Numerical

- Using the following data, compute the annual depreciation and book value for each year by:

1. Straight Line (SL) method

2. Declining Balance (DB) method

3. Sum-of-the-Years'-Digits (SOYD) method

4. Sinking Fund method

Given:

- Cost basis (I) = \$7,000
- Salvage value (S) = \$2,000
- Useful life (N) = 5 years
- MARR (interest rate) = 10%

Numerical

Straight Line (SL) Method

$$D = \frac{I - S}{N} = \frac{7000 - 2000}{5} = 1000 \text{ per year}$$

Year	Depreciation	Book Value
1	1,000	6,000
2	1,000	5,000
3	1,000	4,000
4	1,000	3,000
5	1,000	2,000

Numerical

Declining Balance (DB) Method

$$d = 1 - \left(\frac{S}{I}\right)^{1/N} = 1 - \left(\frac{2000}{7000}\right)^{1/5} \approx 0.222$$

Year	Book Value	Depreciation (0.222 × BV)
1	7,000	1,554
2	5,446	1,209
3	4,237	940.6
4	3,296.4	731.8
5	2,564.56	569
		Total sum = 5004.4

Final Book Value = 7000 - 5004.4 = 1995.6 ≈ 2000

Numerical

Sum-of-the-Years'-Digits (SOYD) Method

$$\text{SOYD} = \frac{N(N+1)}{2} = \frac{5(6)}{2} = 15$$

Total depreciable amount:

$$7000 - 2000 = 5000$$

Year	Depreciation	Book Value
1	$(5/15) \times 5000 = 1,666.67$	5,333.33
2	$(4/15) \times 5000 = 1,333.33$	4,000.00
3	$(3/15) \times 5000 = 1,000.00$	3,000.00
4	$(2/15) \times 5000 = 666.67$	2,333.33
5	$(1/15) \times 5000 = 333.33$	2,000.00

Sinking Fund Method

Sinking Fund Method

$$D = (I - S)(A/F, 10\%, 5)$$

Using factor:

$$(A/F, 10\%, 5) = \frac{0.10}{(1.10)^5 - 1} \approx 0.1638$$

$$D = 5000 \times 0.1638 = 819 \text{ per year (approx)}$$

Book value reduces slowly in early years and faster later because interest accumulates.

Year	Depreciation	Book Value
1	819	6,181
2	901 = 819(1.1)	5,280
3	991 = 901(1.1)	4,289
4	1,090 = 991(1.1)	3,199
5	1,199 = 1,090(1.1)	2,000

Modified Accelerated Cost Recovery System

MACRS is a tax depreciation method used in the United States for income tax purposes. It allows businesses to recover the cost of assets using predefined accelerated percentages over a specified recovery period.

It is not based on salvage value

It uses government-prescribed tables

Key Features

- Accelerated depreciation method
- Salvage value is ignored
- Uses fixed IRS percentage tables
- Different asset classes (3-year, 5-year, 7-year, etc.)
- Based on tax rules, not accounting logic

Modified Accelerated Cost Recovery System

Before 1981, the primary methods used for depreciation were Straight-Line (SL), Declining Balance (DB), and Sum-of-the-Years'-Digits (SOYD).

Between 1981 and 1986, tangible property was depreciated under the Accelerated Cost Recovery System (ACRS). This system was introduced by the Economic Recovery Tax Act of 1981 (ERTA) in the United States.

After 1986, the Tax Reform Act of 1986 (TRA 86) modified the earlier ACRS system and made it mandatory to use the Modified Accelerated Cost Recovery System (MACRS) for depreciating tangible property.

MACRS became the principal method for computing depreciation in engineering and tax-based project evaluations. Under MACRS:

- The salvage value is assumed to be zero.
- Useful life estimates are not directly used to compute depreciation.
- Instead, assets are classified into standard recovery periods.

The MACRS system includes eight asset categories with recovery periods of:

3, 5, 7, 10, 15, 20, 27.5, and 39 years

(as specified by the Internal Revenue Service – IRS).

Modified Accelerated Cost Recovery System

Asset Categories Under MACRS

- 3-year property: Short-lived assets such as plastic products; depreciated using 200% declining balance (DB), then switching to straight-line (SL).
- 5-year property: Computers, automobiles, and light trucks; depreciated using 200% DB, then switching to SL.
- 7–10 year property: Most manufacturing equipment, office furniture, and fixtures; depreciated using 200% DB, followed by a switch to SL.
- 15-year property: Sewage treatment plants and telephone distribution systems; depreciated using 150% DB, then switching to SL.
- 20-year property: Sewer pipes and electrical power plants; depreciated using 150% DB, then switching to SL.
- 27.5-year property: Residential rental property; depreciated using the straight-line method.
- 39-year property: Non-residential real estate (commercial buildings), including elevators and escalators; depreciated using the straight-line method.

Internal Revenue Service (IRS) MACRS percentages table

Year	3-Year	5-Year	7-Year	10-Year	15-Year	20-Year
1	33.33	20.00	14.29	10.00	5.00	3.750
2	44.45	32.00	24.49	18.00	9.50	7.219
3	14.81	19.20	17.49	14.40	8.55	6.677
4	7.41	11.52	12.49	11.52	7.70	6.177
5	—	11.52	8.93	9.22	6.93	5.713
6	—	5.76	8.92	7.37	6.23	5.285
7	—	—	8.93	6.55	5.90	4.888
8	—	—	4.46	6.55	5.90	4.522
9	—	—	—	6.56	5.91	4.462
10	—	—	—	6.55	5.90	4.461
11	—	—	—	3.28	5.90	4.462
12	—	—	—	—	5.91	4.461

Internal Revenue Service (IRS) MACRS percentages table

13	—	—	—	—	5.90	4.462
14	—	—	—	—	5.91	—
15	—	—	—	—	5.90	—
16	—	—	—	—	2.95	—
17	—	—	—	—	—	—
18	—	—	—	—	—	—
19	—	—	—	—	—	—
20	—	—	—	—	—	—
21	—	—	—	—	—	2.231

Important note: In MACRS (tax depreciation system), IRS assumes:

The asset is placed in service in the middle of the year, no matter when you actually bought it.

So:

First year → only half year of operation

Last year → remaining half year

Modified Accelerated Cost Recovery System

- A taxpayer places in service an asset costing Rs. 10,000, classified under the 5-year MACRS property class . Compute the MACRS percentages and the depreciation amount for the asset. Given : Useful life = 5 years, Cost of asset = Rs. 10,000 Salvage value = 0

Straight-Line Rate (SL):

$$\text{SL rate} = \frac{1}{N} = \frac{1}{5}$$

Declining Balance Rate (DB):

$$\text{DB rate} = \frac{R}{N}$$

For 5-year property, IRS guideline specifies:

$$R = 200\%$$

$$\text{DB rate} = \frac{200\%}{5} = 40\%$$

Modified Accelerated Cost Recovery System

Year	MACRS % Calculation	Selected MACRS %
1	$\frac{1}{2}$ Year DB = $\frac{1}{2}(0.40)(100\%) = 20\%$ $\frac{1}{2}$ Year SL = $\frac{1}{2}(1/5)(100\%) = 10\%$	20%
2	DB = $(0.40)(100\% - 20\%) = 32\%$ SL = $(1/4.5)(80\%) = 17.78\%$	32%
3	DB = $(0.40)(100\% - 20\% - 32\%) = 19.2\%$ SL = $(1/3.5)(48\%) = 13.7\%$	19.20%
4	DB = $(0.40)(100\% - 52\% - 19.2\%) = 11.52\%$ SL = $(1/2.5)(28.8\%) = 11.52\%$	11.52%

Modified Accelerated Cost Recovery System

5	$DB = (0.40)(17.28\%) = 6.9\%$	11.52%
	$SL = (1/1.5)(17.28\%) = 11.52\%$	

6	$\frac{1}{2} \text{ Year DB} = \frac{1}{2}(0.40)(5.76\%) = 1.15\%$	5.76%
	$\frac{1}{2} \text{ Year SL} = \frac{1}{2}(1/5)(5.76\%) = 5.76\%$	

IRS assumes

$\frac{1}{2}$ year (Year 1)

1 year

1 year

1 year

1 year

$\frac{1}{2}$ year (Year 6)

So total = 5 full years.

In MACRS (tax depreciation system), IRS assumes:

The asset is placed in service in the middle of the year, no matter when you actually bought it.

So:

First year → only half year of operation

Last year → remaining half year

Modified Accelerated Cost Recovery System

The depreciation amount for the asset.

Year	MACRS %	Depreciation (Rs.)
1	20%	2,000
2	32%	3,200
3	19.20%	1,920
4	11.52%	1,152
5	11.52%	1,152
6	5.76%	576

Tax and corporate income tax

A **tax** is money that individuals or businesses must pay to the government. It is a mandatory financial charge imposed by government that directly affects project cash flows. It is not just a legal concept, it is a cash flow modifier in project evaluation.

A) Direct Taxes

- Paid directly by individuals or corporations to the government.
- Cannot be shifted to someone else.

Examples:

- Personal income tax
- Corporate income tax
- Property tax

In engineering economics, corporate income tax is the most important direct tax because it directly affects project cash flows.

Tax and corporate income tax

(B) Indirect Taxes

- Levied on goods and services.
- Can be shifted to consumers.

Examples:

- VAT
- Sales tax
- Excise duty
- Customs duty

Tax and corporate income tax

The **corporate tax** rate is applied to the taxable income of a corporation. A corporation (company) does not pay tax on total revenue, it pays tax only on taxable income. Before computing corporate tax, we must determine taxable income. Allowable deductions include:

- Cost of goods sold
- Salaries and wages
- Rent
- Interest
- Advertising
- Depreciation
- Amortization
- Depletion

$\text{Taxable Income} = \text{Revenue} - \text{Allowable Deductions}$
Where depreciation is a key deduction.

Tax and corporate income tax

<i>Concept</i>	<i>Meaning</i>	<i>Who It Applies To</i>	<i>How It Is Calculated</i>	<i>Role in Engineering Economics</i>
Corporate Income Tax	Tax paid by a company on its taxable profit	Corporations / Firms	$(T = t(R - E - d))$	Directly affects After-Tax Cash Flow (ATCF) and project evaluation (NPW, IRR)
Personal Income Tax	Tax paid by individuals on their income	Individuals / Employees	Based on salary or personal income tax brackets	Not directly used in project evaluation (unless analyzing personal investment)
Book Value (BV)	Accounting value of an asset at a given time	Business asset records	$(BV = P - \text{Accumulated Depreciation})$	Used to determine gain/loss on asset disposal and tax calculation
Depreciation	Allocation of asset cost over useful life	Business fixed assets	Depends on method (SL, DB, SOYD, MACRS)	Reduces taxable income → Creates tax shield → Increases ATCF

Economic analysis (After-tax cash flow)

After-Tax Economic Analysis

After-tax economic analysis refers to measuring the profitability of a project after considering income taxes. In simple terms, it is the evaluation of after-tax cash flows (ATCFs).

Procedure for After-Tax Economic Analysis

Let:

- R_k = Revenue from the project during period k
- E_k = Expenses during period k
- d_k = Depreciation during period k
- t = Income tax rate (assumed constant over the study period)
- T_k = Income tax amount during period k

Economic analysis (After-tax cash flow)

ATCF (After-Tax Cash Flow) — Formulas

- $BTCF_k = R_k - E_k$ (*Before Tax Cash Flow*)
- $\text{Taxable Income}_k = R_k - E_k - d_k$
- $T_k = t(R_k - E_k - d_k)$
- $ATCF_k = BTCF_k - T_k$
- $ATCF_k = R_k - E_k - t(R_k - E_k - d_k)$
- $ATCF_k = (1 - t)(R_k - E_k) + td_k$
- $\text{Tax Shield}_k = td_k$

Economic analysis (After-tax cash flow)

- An asset has an installed cost of Rs. 45,000 with a salvage value $S_v=0$. It is classified as a 5-year property under MACRS.

Determine the appropriate MACRS depreciation schedule.

The asset is expected to generate annual revenue of Rs. 23,000 for 6 years, with an annual operating cost of Rs. 7,300. The required rate of return is 15% (after tax), and the income tax rate is 40%.

Evaluate the after-tax Internal Rate of Return (IRR) using the annual worth method.

Economic analysis (After-tax cash flow)

Year	BTCF = Revenue – Cost	MACRS Depreciation (d_k)	Taxable Income (BTCF – d_k)	Tax @ 40%	ATCF = BTCF – Tax
0	–45,000	—	—	—	–45,000
1	$\frac{1}{2}(23,000 - 7,300) = 7,850$	20% of 45,000 = 9,000	$7,850 - 9,000 = -1,150$	No tax	7,850
2	15,700	32% of 45,000 = 14,400	1,300	$40\% \times 1,300 = 520$	15,180
3	15,700	19.2% of 45,000 = 8,640	7,060	2,824	12,876
4	15,700	11.52% of 45,000 = 5,184	10,516	4,206.4	11,493.6
5	15,700	11.52% of 45,000 = 5,184	10,516	4,206.4	11,493.6
6	7,850	5.76% of 45,000 = 2,592	$7,850 - 2,592 = 5,258$	2,103.2	5,746.8

Economic analysis (After-tax cash flow)

Present Worth at 10%

$$\begin{aligned}PW(10\%) &= -45,000 + 7,850(1.1)^{-1} + 15,180(1.1)^{-2} \\&\quad + 12,876(1.1)^{-3} + 11,493.6(1.1)^{-4} + 11,493.6(1.1)^{-5} + 5,747(1.1)^{-6} \\&= -45,000 + 7,136.36 + 12,545.45 + 9,674 + 7,850 + 7,136.6 + 3,244 \\&= -45,000 + 47,586.4 = 2,586.4\end{aligned}$$

Annual Worth at 10%

$$\begin{aligned}AW(10\%) &= 2,586.4(A/P, 10\%, 6) \\&= 2,586.4(0.2296) = 593.8\end{aligned}$$

Economic analysis (After-tax cash flow)

Present Worth at 15%

$$\begin{aligned}PW(15\%) &= -45,000 + 7,850(1.15)^{-1} + 15,180(1.15)^{-2} \\&+ 12,876(1.15)^{-3} + 11,493.6(1.15)^{-4} + 11,493.6(1.15)^{-5} + 5,747(1.15)^{-6} \\&= -45,000 + 6,826 + 11,478 + 8,466 + 6,571 + 5,714 + 2,485 \\&= -3,433\end{aligned}$$

Annual Worth at 15%

$$\begin{aligned}AW(15\%) &= -3,433(A/P, 15\%, 6) \\&= -3,433(0.2642) = -906\end{aligned}$$

Economic analysis (After-tax cash flow)

IRR by Annual Worth Interpolation

$$\begin{aligned} IRR &= 10\% + \frac{593.8}{593.8 + 906}(15\% - 10\%) \\ &= 10\% + 1.98\% \end{aligned}$$

$$IRR \approx 11.98\%$$

Here: $11.98\% < 15\%$

So, the project is NOT feasible (reject).

Assignment

- 1) Compute the annual depreciation allowances and the resulting book value using the double declining balance method with switching to the straight-line method.
 - Cost of asset = Rs. 1,00,000
 - Useful life = 5 years
 - Salvage value = Rs. 2,000
- 2) A company purchased a machine for Rs. 25,000, expected to generate a benefit of Rs. 8,000 per year for five years. The salvage value at the end of five years is Rs. 10,000. Calculate the after-tax cash flow, given: Tax rate = 40% Depreciation method = Sinking Fund Method , Interest rate = 20%
- 3) A machine is expected to cost Rs. 5,00,000 and will generate revenue of Rs. 1,50,000 per year for five years. Its salvage value is Rs. 2,00,000.
Calculate after-tax cash flow and corresponding NPV if:
Tax rate = 30%
Depreciation method = Sum of Year Digits (SOYD)
MARR = 15%

Assignment

4) Using the following data, compute the annual depreciation and book value for each year by:

- (i) Straight Line (SL) method
- (ii) Declining Balance (DB) method
- (iii) Sum-of-the-Years'-Digits (SOYD) method
- (iv) Sinking Fund method

Data:

- Cost basis = \$7,000
- Salvage value = \$2,000
- Useful life = 5 years
- MARR = 10%

Assignment

- 5) What do you mean by depreciation and what are its causes? A machine purchased for Rs. 50,000 by expecting useful life of 10 years. Calculate depreciation amount for each year by using declining balance method when rate of depreciation is 20% per year.
- 6) A machine was purchased for Rs. 50,000 with an expected useful life of 10 years. Calculate the annual depreciation amount for each year using the declining balance method when the rate of depreciation is 20% per year.

Thank you for your attention!